distortion and warping, since the object is printed in layers, which involves change in temperatures in different parts of the object.

3D printed antennas

Most antennas require a conductive material to provide power when transmitting and receiving signals. Therefore, metal 3D printers seem like a viable option. Antennas are extensively used in the aerospace industry in all commercial and military aircraft, satellites, etc. Metal 3D printing can allow manufacturers to design and develop intricate antenna shapes with less weight and at a lower cost.

Metallic printing has been used to fabricate horn antennas, waveguides, and other types of antennas. This technique has been known to drastically reduce the number of components required in an antenna, thus driving down cost, size, weight, and complexity.

A company named Optisys was able to reduce the number of parts in the assembly from over one hundred to just a single piece using metal 3D printing. This helped the company to reduce the lead time from eleven months to two months, while saving over 20% in the production cost.

One downside of fabricating antennas using a 3D printer is that these antennas have rough surfaces, which can affect the quality of signals. Therefore, the antennas need to go through extensive post-processing before they can be used in commercial applications.

In the graph on previous page you can see how the smooth surface antenna shows considerably lower return loss compared to a rough surfaced antenna.

Conclusion

Metal 3D printing is an additive manufacturing technique that opens new realms to how we manufacture and fabricate metallic components. This technique enables us to create intricate objects while also reducing the time required for the proof of concept phase when making prototypes.

Metal 3D printing can be used in a wide variety of industries including automotive, electronics, space, and defence. The use of metal 3D printing in designing antennas can change how we design and manufacture antennas for communication applications. EFY

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